

CampusHub

An College Management System

Project Synopsis Presentation



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Introduction

Building a secure, scalable web-based College Management System to manage students, faculty, courses, attendance, examinations, results, fees, library, timetable, and reporting.

Problem Statement - CampusHub solves the real-world problem of inefficient and disconnected college administration by centralizing student, faculty, academic, and administrative activities into a single digital platform.

Scope

User (role based access, secure login and authentication), Student & Faculty Management (admissions, registrations, profile management, etc.), Academic (curriculum and course planning), Attendance, Examinations & Assessment, Fee, Timetable, Library management, etc.

Out of Scope: Online classes, biometrics, online exams, mobile app development, etc.

Feasibility

Technical Feasibility - Technically feasible with current tools, resources, and expertise.

Economic Feasibility - Economically feasible with potential for substantial ROI (Return of Investment).

Operational Feasibility - Operationally feasible with user-friendly design and scalable deployment.

Legal & Ethical Feasibility - Feasible with adherence to ethical AI practices and data protection laws.

Methodology

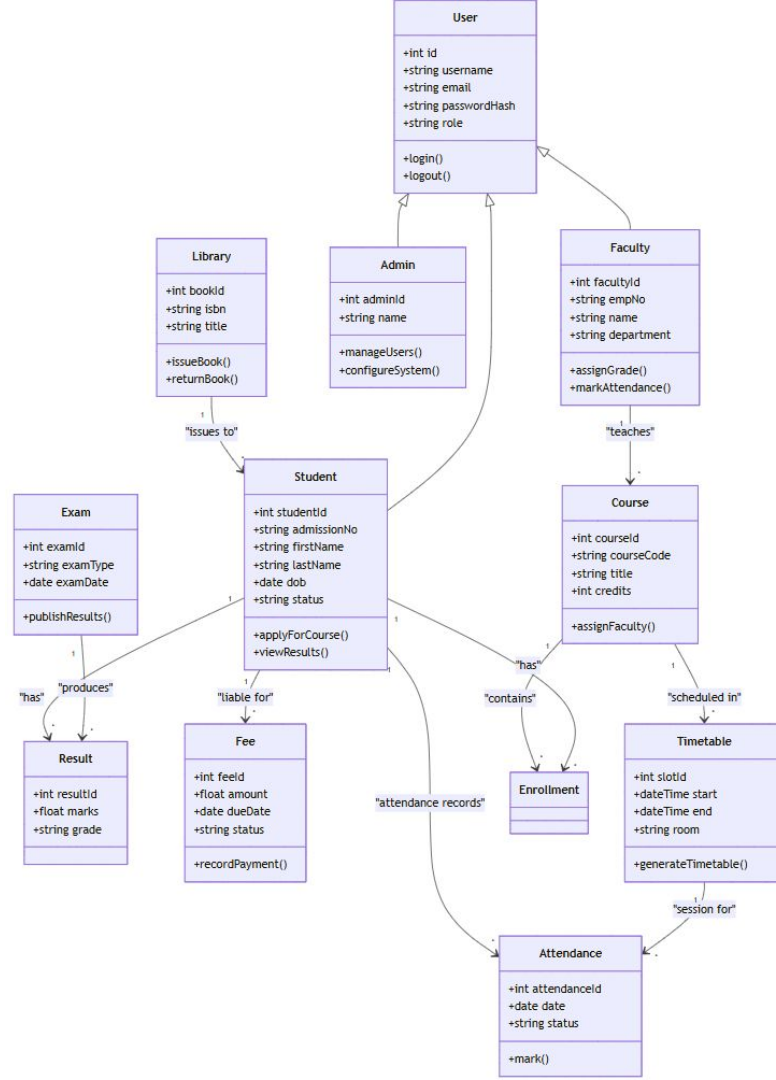
1. **Requirement & Planning** - Gather functional and non-functional requirements for college management. Identify user roles, modules, workflows, and system expectations.
2. **System Design** - Create ER diagrams, class diagram, data flow diagram. Technologies, tools, and frameworks for frontend, backend, database, and deployment are finalized.
3. **Frontend Development** - Build a responsive UI in React using component-based architecture. Integrate APIs, implement role-based dashboards, forms, reports, and visualizations
4. **Backend Development** - Develop REST APIs using Spring Boot. Implement modules for auth., students, faculty, attendance, exams, and payments. Use layered architecture and integrate MySQL.
5. **Database Creation** - Design normalized MySQL database schemas, define relationships, constraints, and prepare migration scripts. Ensure data integrity, security, and optimization.
6. **Deployment** - Containerized Deployment (Docker + CI/CD).
7. **Hosting** - 2 options: i) render ii) college servers.

Modules

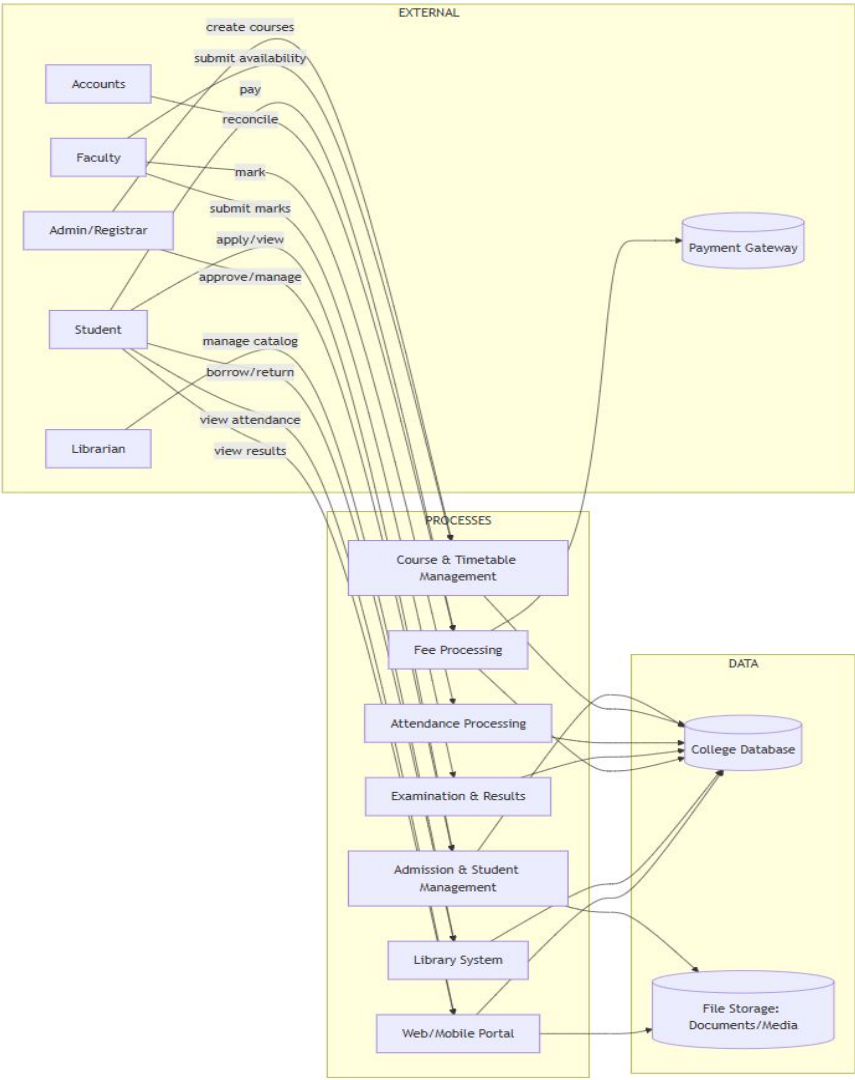
12 Modules

- User & Role Management Module
- Attendance Management Module
- Result/Grade Management Module
- Courses & Subjects Management Module
- Announcements Management Module
- Document Management Module
- Timetable Management Module
- Exam Management Module
- Library Management Module
- Placement Management Module
- Hostel Management Module
- Fee & Finance Management Module

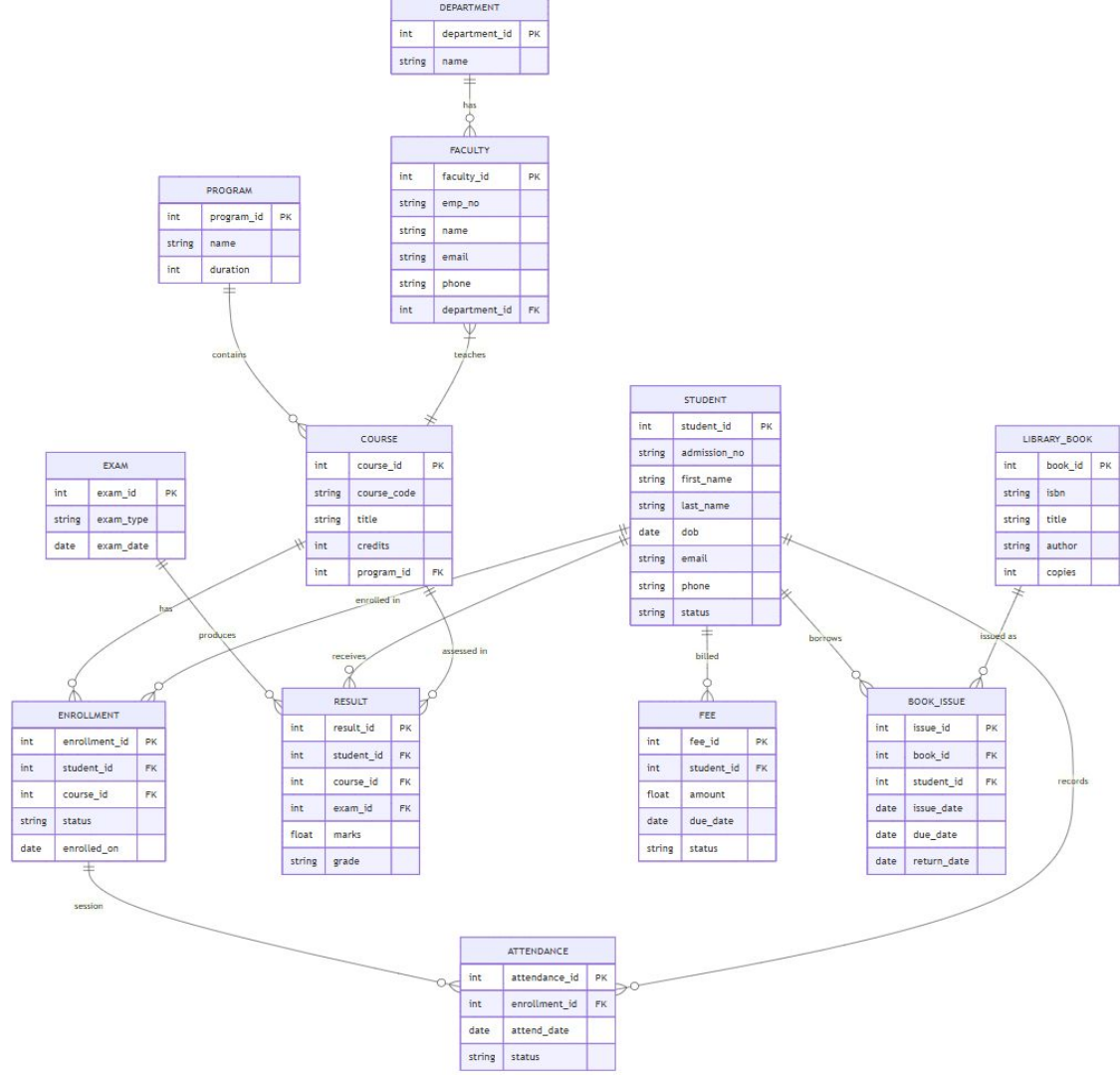
Class Diagram



Data Flow Diagram



ER Diagram



Requirements

Hardware

- Processor: Intel i3 or higher
- RAM: 8 GB minimum
- Storage: 20 GB free space
- Display: 1024x768 resolution minimum

Software

- Operating System - Windows
- Frontend - React.js, HTML, CSS, JavaScript
- Backend - Java Spring Boot
- Database - MySQL
- Tools - VS Code, Eclipse IDE, Postman, Docker
- Version Control - Git, GitHub

Security Issues

- Authentication & Authorization Risks
- Data Security & Privacy
- API Security Issues
- Database & Server Risks
- Deployment & DevOps Issues
- Payment Gateway Issues

Conclusion/Expected Outcome

- A fully functional web-based system for managing college where several users (students, faculties, etc.) can perform different tasks.
- Role-based dashboards for admin, faculty, students, and parents.
- Secure and optimized backend with REST APIs.
- Responsive and user-friendly frontend interface.
- Centralized database with accurate and reliable records.
- Dockerized deployment ready for hosting.

Thank You

a presentation by :- Shikhar Sharma